Tactix: Proof-State and Tier-D Protocol Checkpoint

Report to Andrew

25 September 2026

Outcome

N1–N4 and the cumulative gates remain open. No Tier-D comparison was submitted. The active AM427 Sweep candidate has clean matched Tier A/B results; Tier C is still running. A source/protocol audit identifies two distinct N2 and comparison risks: the canonical H2 DSR checker retains implied-unit reasons after Tactix has deleted them, and the older Tier-D launcher forces DRAT instead of measuring Tactix's automatic certificate path. The technical record is AM432.

Verified evidence and limits

Tier A: both arms solved 199/200 with 117 independently checked UNSAT proofs each, and matching paired verdicts and proof sizes. Tier B: both arms solved 28/28, with every proof and model checked and matching paired verdicts/proof sizes. At 19:20 UTC, Tier C array `1723299` held 55/60 result records: 27 complete pairs were checker-clean and matched in verdict and proof size; five heavy tasks remained live, and dependent aggregate `1723300` had not run. The current-source Debug CTest job `1723391` remained queued. These are not final Tier-C or N4 results.

The H2 current-source 21-variable proof is byte-identical to the historical parent. Frozen no-option `dsr-trim` rejects it backward and forward; the same proof verifies with `-delete-implied-units` through frozen Trestle. Checker source review explains the mismatch, but the frozen-binary replay is the authority. Neither a fixed deletion order nor publishing all root units is a general repair. The option changes proof-state semantics and needs an explicit contract decision plus broad positive and negative audits; the no-option route remains canonical meanwhile. Public action tests reach all 29 registry entries, but that is not proof of every legal N1 boundary.

Tier D and next work

The first-winner authority contains 1,166 unique SC24–26 instances. The requested per-task contract is four CPUs, 32 GiB, 5,000 solver seconds and 45,000 checker seconds on `cpu-dense-hwulab-q`. Its old 23-task throttle was a scratch-management measure; plan §8.1 now requires full-width, unthrottled arrays with anonymous node-local scratch. The dedicated `tactix_tierd.py` currently forces `-drat`. An existing `runtime_task.py` route already supports automatic PBP, witnessed DSR, and witness-free DSR with original-CNF checking; adapt and pin that route before the post-goal comparison. Finish Tier C and current Debug/fuzz validation, resolve H2 generally, then close the retained coverage, performance, rule-parity and N1–N4 gates. The online-learning architecture assessment in the prior AM430 report remains unchanged: production learning needs shared slot-control, owned switchable sub-policies, and live policy session ownership.